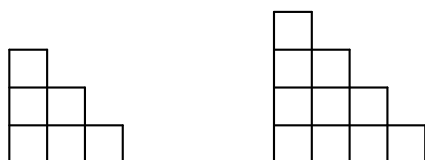


Problem 41. The two figures depicted below consisting of 6 and 10 unit squares, respectively, are called *staircases*.



Consider a 2018×2018 board consisting of 2018^2 cells, each being a unit square. Two arbitrary cells were removed from the same row of the board. Prove that the rest of the board cannot be cut (along the cell borders) into staircases (possibly rotated).

Problem 42. Let ABC be an acute-angled triangle with $AB < AC$, and let D be the foot of its altitude from A . Let R and Q be the centroids of the triangles ABD and ACD , respectively. Let P be a point on the line segment BC such that $P \neq D$ and the points P, Q, R and D are concyclic. Prove that the lines AP, BQ and CR are concurrent.

Problem 43. Let n be a positive integer and let $a_1, \dots, a_k$, where $k \geq 2$, be distinct integers in the set $\{1, \dots, n\}$ such that n divides $a_i(a_{i+1} - 1)$ for $i = 1, \dots, k - 1$. Prove that n does not divide $a_k(a_1 - 1)$.

Problem 44. Determine all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that the equality

$$f(\lfloor x \rfloor y) = f(x) \lfloor f(y) \rfloor$$

holds for all $x, y \in \mathbb{R}$. (Here $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z .)